

DATA STORYTELLING & STATISTICAL VALIDATION

Customer Value & Repeat-Purchase Intelligence

APEXPLANET INTERNSHIP — TASK 4

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BUSINESS PROBLEM & DATA OVERVIEW

Objective: Understand customer value, purchasing behaviour, and repeat purchasing.

KEY BUSINESS QUESTIONS

- Who contributes most to revenue?
- How strong is customer retention?
- Which products and categories perform best?
- Which locations contribute most?
- Is the repeat rate below the business benchmark?

DATASET SUMMARY

992 ORDERS

941 CUSTOMERS

15 COLUMNS
covering Customer, Order,
Product, Category, Quantity,
Sales, Demographics

Task 4 builds on the analytical foundation established in **Tasks 1, 2 and 3.**

EXECUTIVE KPI SNAPSHOT

Key performance indicators at a glance —
driving revenue, customers, and repeat value.

₹138.1M

TOTAL REVENUE

992

TOTAL ORDERS

941

UNIQUE CUSTOMERS

₹139,218

AVG. ORDER VALUE

₹146,763

REVENUE / CUSTOMER

50

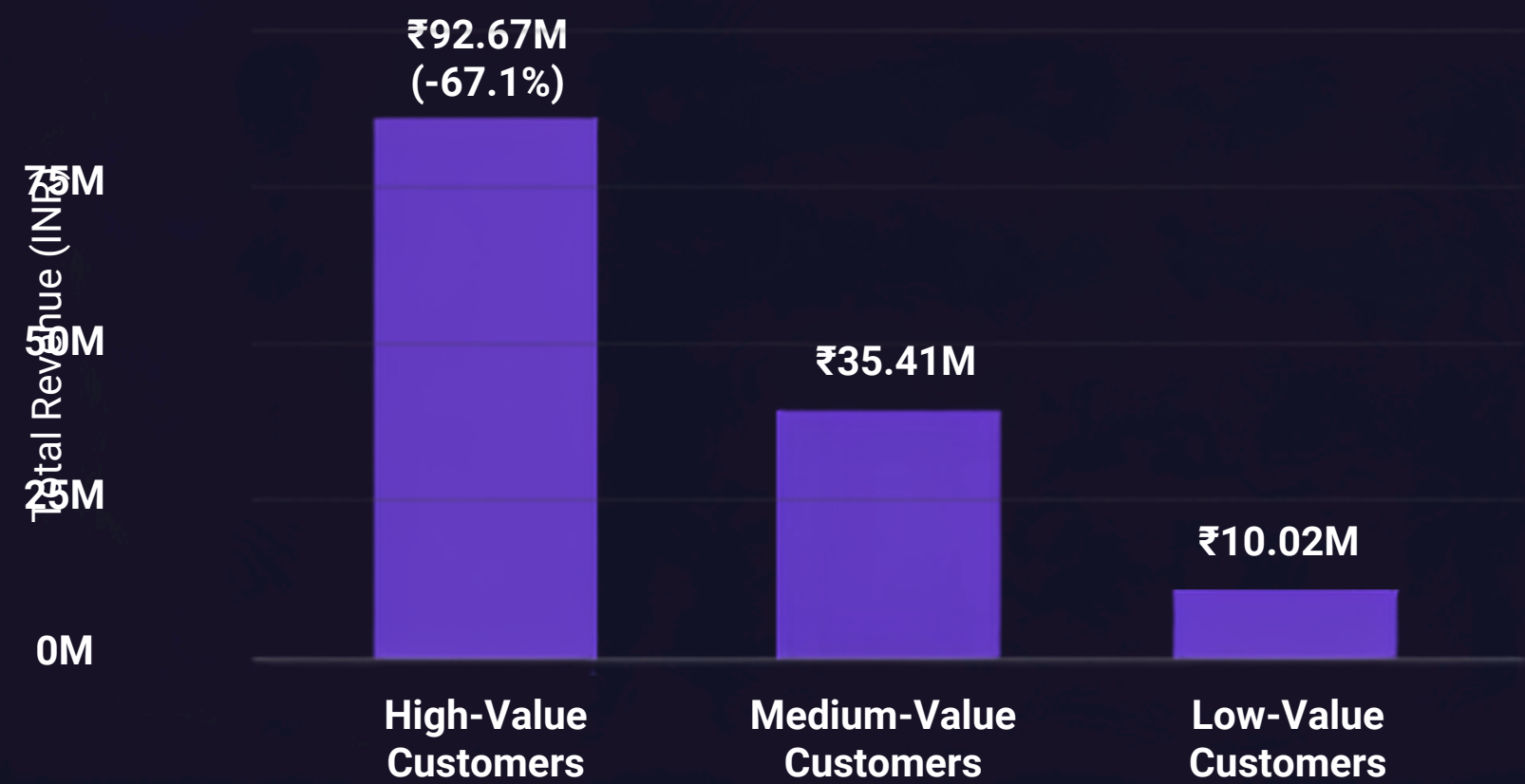
REPEAT CUSTOMERS

5.3%

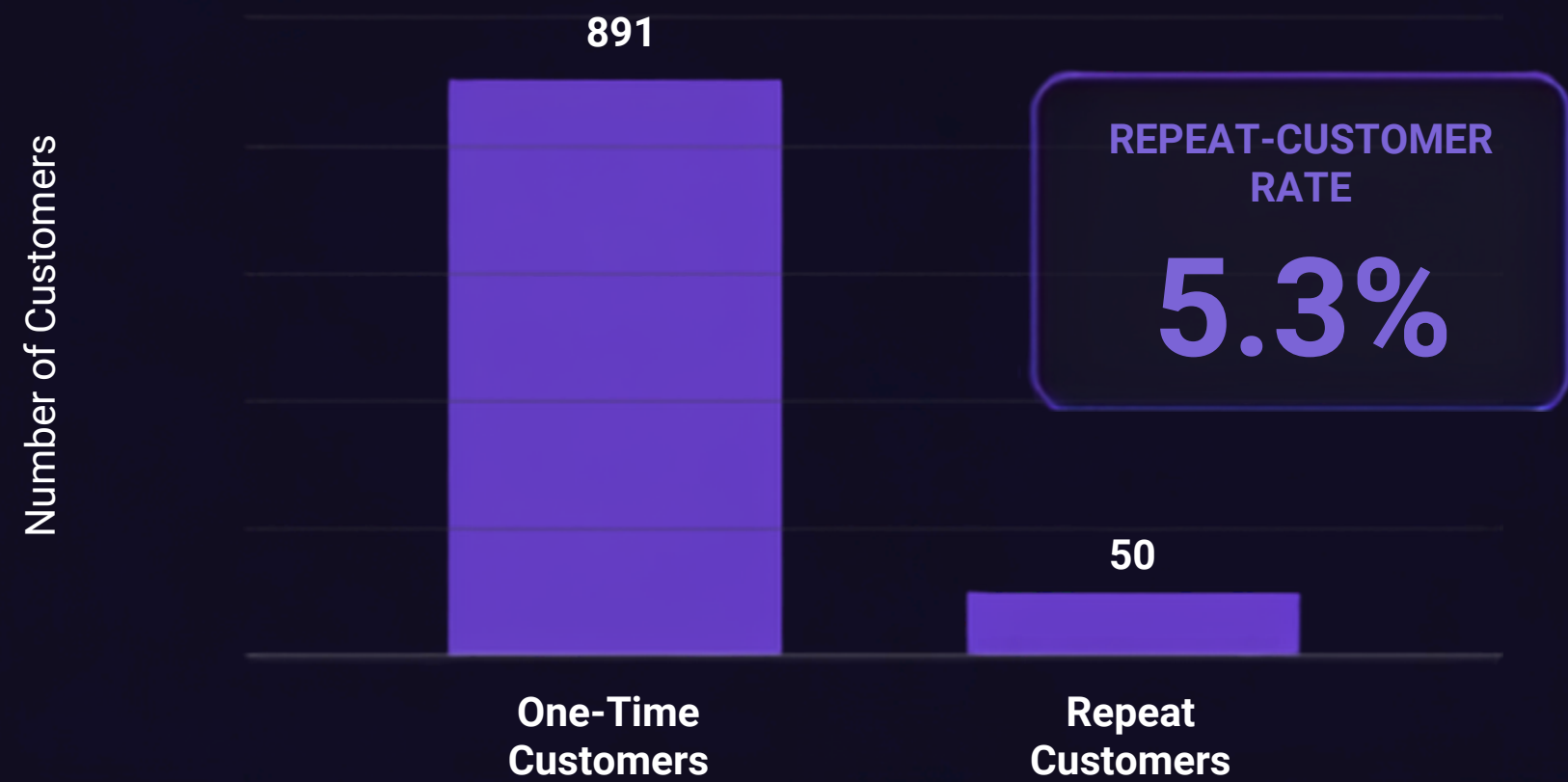
REPEAT RATE

CUSTOMER VALUE & RETENTION

CUSTOMER VALUE SEGMENT



CUSTOMER TYPE



KEY INSIGHT

High-value customers generate approximately two-thirds of revenue, while repeat purchasing remains low.

BUSINESS PRIORITY

Retain high-value customers and increase repeat purchasing.

PRODUCT, CATEGORY & CITY PERFORMANCE

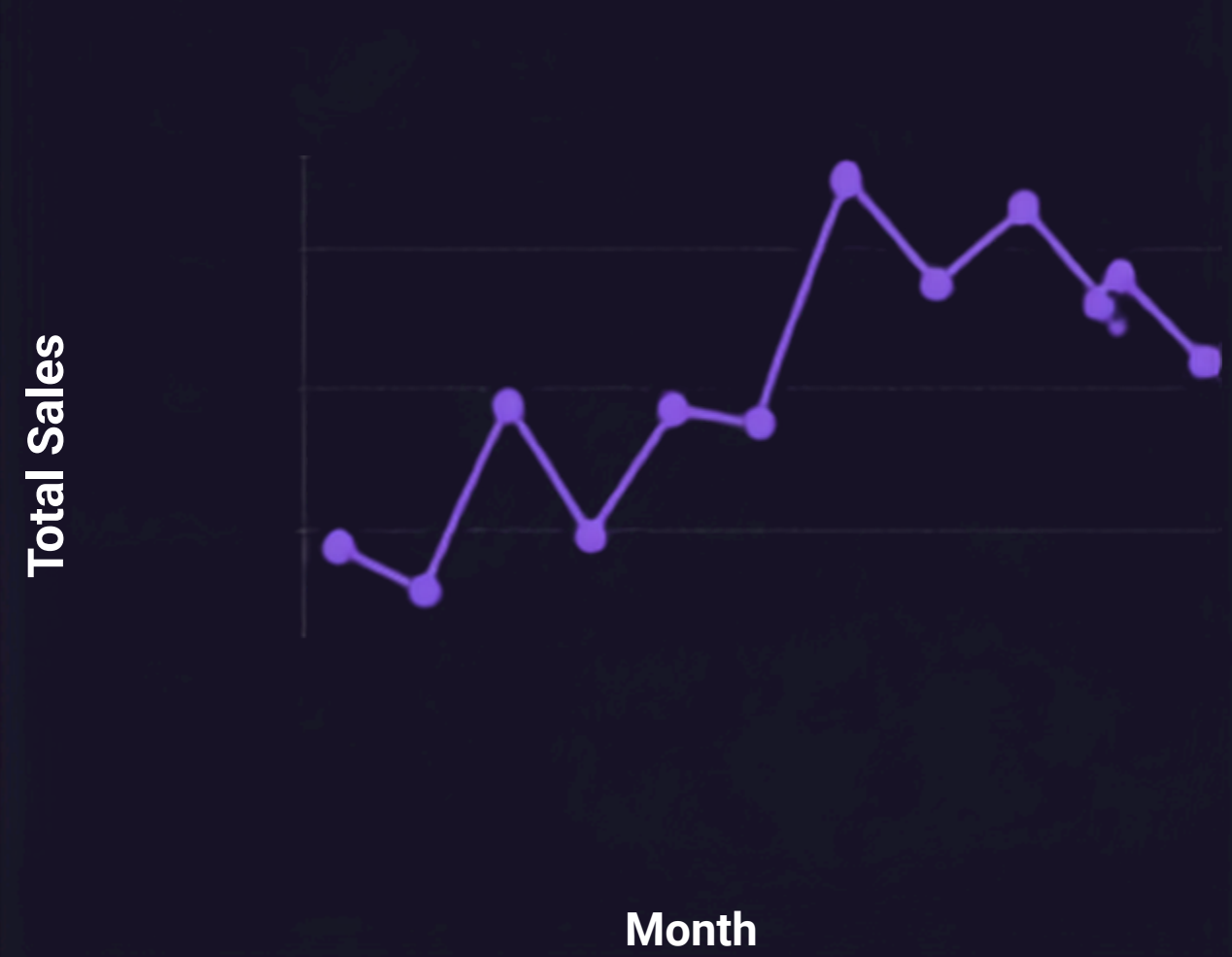
Product, category and geographic performance identify where targeted sales strategies can have the greatest impact.



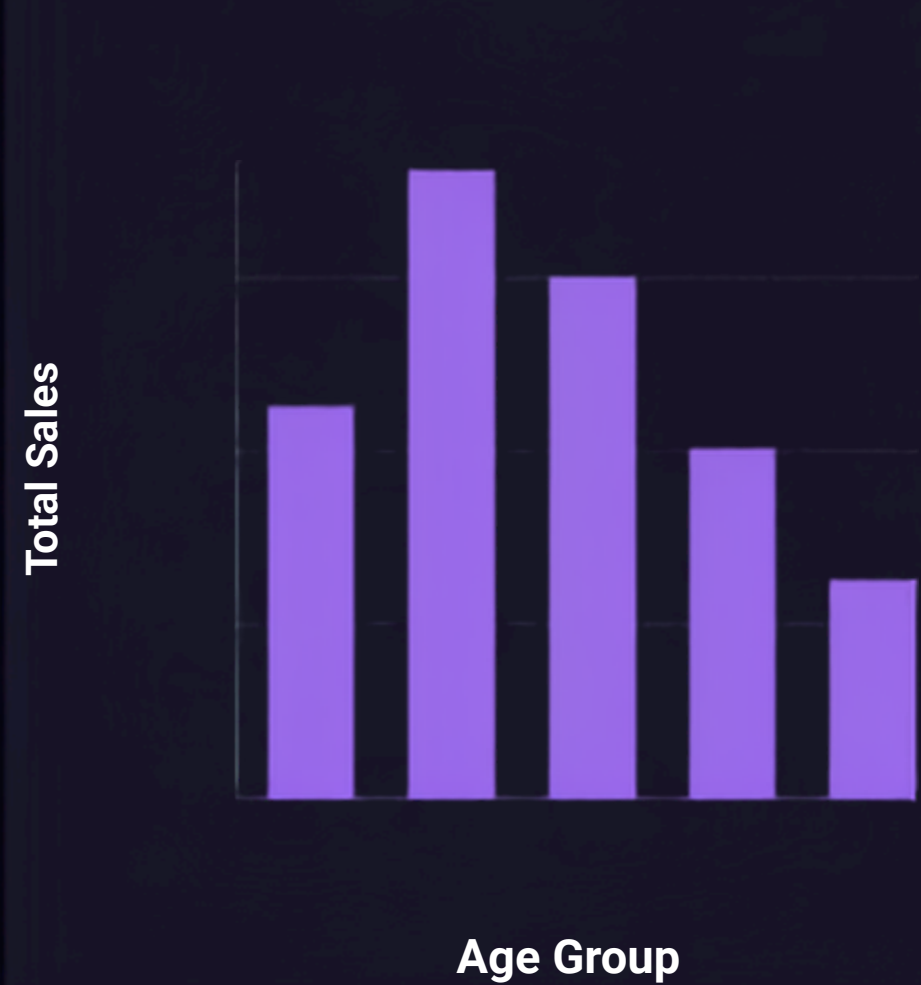
REVENUE & CUSTOMER BEHAVIOUR

Customer Value & Repeat-Purchase Behaviour

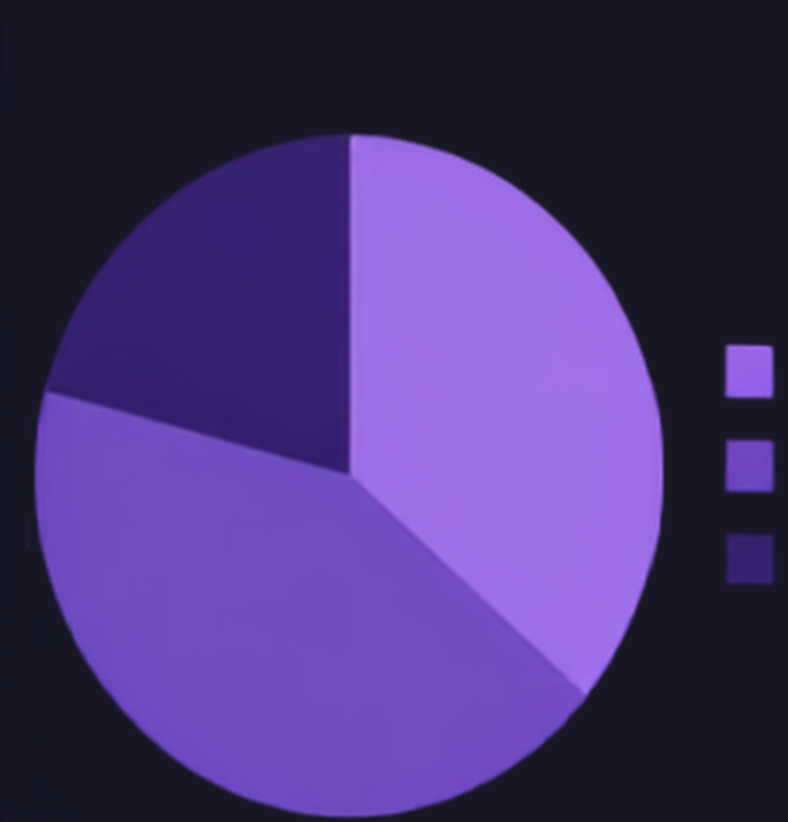
MONTHLY SALES TREND



SALES BY AGE GROUP



REPEAT CUSTOMER ANALYSIS



01

Revenue fluctuates

Sales vary across months, with mid-year and Q3 peaks.

02

Behaviour by age differs

The 26-35 age group drives the highest sales.

03

Repeat purchases are strong

Most customers purchase 2-5 times, showing repeat engagement.

BUSINESS HYPOTHESIS

Q: Is the observed repeat-customer rate significantly below the 10% business benchmark?

H₀

p = 0.10

H₁

p < 0.10

STATISTICAL TEST

One sample proportion z-test

SIGNIFICANCE LEVEL

a = 0.05

OBSERVED REPEAT RATE

5.31%

**10% = SELECTED BUSINESS BENCHMARK
(NOT CALCULATED FROM THE DATASET)**

STATISTICAL VALIDATION & BUSINESS MEANING

OBSERVED RATE	BENCHMARK	Z-STATISTIC	P-VALUE	SIGNIFICANCE LEVEL	DECISION
5.31%	10%	-4.792	0.00000083	0.05	REJECT H_0
repeat-customer rate	business benchmark	test statistic	probability result	threshold	statistical decision

CONCLUSION

The observed repeat-customer rate is significantly below the selected 10% business benchmark.

STATISTICAL DECISION

p-value < 0.05 → REJECT H_0

BUSINESS MEANING

Customer retention is an important opportunity for the business.

STATISTICAL QUALIFICATION

The test establishes that the repeat-customer rate is below the selected benchmark; it does not explain why customers do not return.

ACTIONABLE RECOMMENDATIONS & KEY TAKEAWAYS

ACTIONABLE RECOMMENDATIONS

- 1 Introduce loyalty and repeat-purchase campaigns
- 2 Prioritize retention of high-value customers
- 3 Personalize offers for one-time customers
- 4 Promote strong-performing products and categories
- 5 Target campaigns using city-level performance
- 6 Monitor repeat rate, AOV, and customer value

KEY TAKEAWAYS

₹138.1M TOTAL REVENUE

67.1% FROM HIGH-VALUE CUSTOMERS

5.3% REPEAT CUSTOMER RATE

LAPTOP / ELECTRONICS / PATNA

REPEAT RATE IS SIGNIFICANTLY BELOW 10%

RETENTION IS THE MAJOR OPPORTUNITY

THANK YOU

Questions & Discussion